1. which type of hormone excrete pineal gland_____ ***melatonin***
2. which hormone on flight-or-flight _____ ***cetocholmines.***
3. All steroid hormones are synthesized from _____ ***cholesterol.***
4. Cardiac muscle cells (myocyte) are _____ ***mononucleate.***
5. Cell bodies are located in ventral horn of _____of spinal cord. ***gray matter***
6. _____are involved in Ca2+ sequestering activity. ***Long longitudinal tubules***
7. In a relaxed muscle fiber, _____blocks crossbridging b/w myosin and actin filaments. ***Tropomyosin***
8. Each quantum may consist of about _____molecules of transmitter molecules. ***2000 to 10,000***
9. The G proteins family consists of about _____ different proteins that are composed of three subunits: α, β and γ. ***20***
10. are composed of about 300 myosin molecules. ***Thick Filaments***
11. Thick Filaments are about 16 nm in diameter. ***16 nm***
12. Thin filaments also have a three polypeptide complex troponin at intervals of about _____ along the thin filament. ***40nm***
13. Synaptic Cleft is A narrow, fluid-filled space, about wide. ***20 nm***
14. Thin filaments are _____ thick and extend across the I band. ***7-8 nm***
15. Troponin complex binds to tropomyosin about every _____ along the thin actin filament. ***40 nm***
16. Terminal cisternae are larger chambers that are closely associated with T tubules. ***(12 nm apart)***
17. In Aspartate (Aspartic Acid) the Excitatory transmitter, primarily localized to the_____ ***ventral spinal cord.*** Also with Produced in the ***mitochondria.***
18. Main nutritional part of the nerve cell_____ ***Cell Body***
19. Voltage-Gated Na+ Channels that is the Fast-acting channels. Activated by _____ and produce rising phase of action potential. ***Depolarization***
20. _____has nondiffusible peptides and proteins that have many carboxyl and other anionic groups. ***Cytosol***
21. **Acclimation** refers to the same process as acclimatization, but the changes are induced Experimentally in the laboratory or field by the investigator.
22. **hypothalamus** sends appropriate messages to the **effector organs**
23. **Facilitated transport** is a passive process that does not require energy in the Form of **ATP.**
24. The active transport of solutes is carried out by specialized **carrier proteins** which can move solutes Across the plasma membrane against their concentration gradients.
25. The opening of voltage-gated Na+Channels in response to **depolarization**
26. Receptors for thyroid Hormones are found in The **nucleus.**
27. Steroid hormones have **Two receptors in a cell.**
28. **Melatonin** causes a Decrease in the secretion Of **gonadotropic** Hormones from pituitary,
29. **Pineal gland** It is linked to the Light-sensing organ, the Parietal eye or third eye
30. Hormonal activity links the **two cycles.**
31. The glucocorticoids, like Other steroid hormones, Bind to specific receptors In the **cytosol**, forming **Hormone-receptor Complexes.**
32. Corticosteroids belong to **Three functional Categories**

**33.** There are two groups of Adrenoreceptors. **α & β.**

**34.** AM + ATP = **A + M-ATP**

35. **Cell body** is the main nutritional part of nerve cell

***36.*** Donnan Equilibrium Discovered by _____ ***Frederick Donnan in 1911.***

***37.*** _____Changes are induced experimentally in the laboratory. ***Acclimation***

***38.*** Principle of Homeostasis given by _____ ***Walter Cannon (1929)***

***39.*** The rate of diffusion of a solute is calculated by the_____ ***Fick Diffusion Equation.***

***40.*** The effect of single ion gradient on membrane potential is predicted by the _____ ***Nernst equation.***

***41.*** By the mid-1960s, only _____ neurotransmitters had been identified. ***Three***

***42.*** Choline is actively reabsorbed by the _____ ***presynaptic membrane***

***43.*** Neurosecretory peptides are liberated into the circulation and carried by the blood to their_____ ***target neurons.***

***44.*** The tension produced by the muscle is maximum when _____of cross-bridges are formed between actin and myosin. ***largest number***

***45.*** The force for pulling is produced by the _____ of myosin heads. ***partial rotation.***

***46.*** Amine hormones (thyroid and adrenal medullary hormones) are synthesized by the _____ on amino acid tyrosine. ***action of enzymes***

***47.*** Newly synthesized thyroid hormones are incorporated into _____ of the protein thyroglobulin. ***Macromolecules.***

***48.*** Antidiuretic Hormone (ADH) also known as ***Vasopressin and Oxytocin,*** These hormones are synthesized by the cell bodies of neurons in the _____ ***hypothalamus.***

***49.*** Adrenocorticotropic Hormone (ACTH), ACTH is a peptide hormone comprising single chain of _____amino acids. ***39***

***50.*** _____Its principal function is to regulate secretion of corticosteroid hormones by the cortex of adrenal gland. ***Adrenocorticotropic Hormone (ACTH)***

***51.*** Calcitonin is A protein hormone secreted by the _____ cells of thyroid gland. ***Parafollicular***

***52.*** Release of epinephrine and norepinephrine is controlled by the action of _____ sympathetic nerves. ***Preganglionic.***

***53.*** Glucocorticoid secretion is regulated by _____ by the hormones themselves. ***negative feedback.***

***54.*** Increasing rate of uptake of glucose by the cells of _____ ***liver, muscles and adipose tissue.***

***55.*** Steroid Sex Hormones: They are produced by the_____ ***gonads ovaries and testes.***

***56.*** Intracellular conc. of Na+ & Clis maintained about ten times lower than the _____***extracellular conc***

***57.*** Androgens stimulate spermatogenesis by the germinal cells of _____***testes.***

***58.*** As a result _____ are the most conc. inorganic ions in the cytosol. ***K+***

***59.*** Which ion is found in high concentration_____***Sodium, potassium, and chloride***

***60.*** The conc. of K+ is maintained _____more in the cytosol than extracellular fluid. ***10-30 times***

***61.*** Acetylcholine: ACh is excitatory at the neuromuscular junctions It opens channels that allow ***Na+ to flow in and K+ to flow out.***

62. diffusion is process of ***passive transport***.

***63.*** Animal Models for Human Diseases: 1= Diabetic mice 2= Congenitally fat rats 3= Zebra fish embryos with heart defects 4=Provide insight into the physiological processes of disease. Which is not part of them is ***Alzheimer's disease***

64. Adrenal cortex produces ***two*** major types of steroid hormones: • Mineralocorticoids • Glucocorticoids.

***65.*** Based on their structure and pathway for synthesis, hormones are divided into _____ groups. ***Four***

***66.*** Pituitary Gland Also known as _____***hypophysis.***

***67.*** Ionic pumps that produce net charge movement are known as _____***rheogenic.***

***68.*** The opening of voltage-gated Na+ channels in response to depolarization, and the resulting flow of Na+ ions into the cell, is known as _____***membrane excitation.***

69. Voltage-Gated K+ Channels are Known as _____"***delayed rectifiers***"

***70.*** Impulses jump over node to node, Such impulses are known as _____ ***saltatory impulses.***

71. potential change at receptor site is known as_____ "***receptor potential***".

***72.*** The flow of ions that causes an EPSP is known as_____ ***EPSC.***

73. The Muscle Cells: The membrane of the cell is called ***sarcolemma***. Its cytoplasm is called ***sarcoplasm***. Its endoplasmic reticulum is known as ***sarcoplasmic reticulum.***

***74.*** Membrane potential of excited muscle fiber is known as _____***end-plate potential.***

***75.*** The sequence of events that convert an action potential to muscle contraction is known as _____ ***excitationcontraction coupling.***

***76.*** A sustained muscle contraction evoked by stimulation from simultaneous multiple impulses is known as _____ ***tetanic contraction.***

***77.*** Antidiuretic Hormone (ADH) also known as _____***Vasopressin.***

***78.*** Tetraiodothyronine (T4 ) also known as _____ ***thyroxine***

***79.*** Parafollicular cells which secrete one hormone _____ ***Calcitonin***

***80.*** GIP, also known as _____***glucose-dependent insulin-releasing peptide***